

Linear Transformations I

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1 Linear Transformation

1.1 Motivation: Matrices as Functions

Let F be a field and $M_{m \times n}(F)$ be the collection of all $m \times n$ matrices over the field F . Let $A \in M_{m \times n}(F)$ be such a matrix. Every matrix naturally defines a function

$$f_A : F^n \rightarrow F^m \text{ defined by } f_A(v) = Av.$$

By using properties of matrix multiplication, we have

- $f_A(\mathbf{v}_1 + \mathbf{v}_2) = A(\mathbf{v}_1 + \mathbf{v}_2) = A\mathbf{v}_1 + A\mathbf{v}_2 = f_A(\mathbf{v}_1) + f_A(\mathbf{v}_2)$ for all $\mathbf{v}_1, \mathbf{v}_2 \in F^n$.
- $f_A(\lambda \mathbf{v}) = A(\lambda \mathbf{v}) = \lambda(A\mathbf{v}) = \lambda f_A(\mathbf{v})$ for every $\lambda \in F$ and $\mathbf{v} \in F^n$.

Our goal is to study functions on *other vector spaces* that *behave like matrices*. We call them *linear transformations*.

1.2 Definition

Matrices can be viewed as functions between vector spaces. This motivates the following definition.

Definition 4.2.1. Let V and W be vector spaces over the same field F . A function $f : V \rightarrow W$ is called a **linear transformation/linear map/linear function** if it satisfies:

1. (Additivity) $f(\mathbf{v}_1 + \mathbf{v}_2) = f(\mathbf{v}_1) + f(\mathbf{v}_2), \quad \forall \mathbf{v}_1, \mathbf{v}_2 \in V.$
2. (Homogeneity) $f(\lambda \mathbf{v}) = \lambda f(\mathbf{v}), \quad \forall \lambda \in F, \mathbf{v} \in V.$

1.3 Theorem

Theorem 4.3.1. Suppose $f : V \rightarrow W$ be a linear transformation. Then $f(\mathbf{0}_V) = \mathbf{0}_W$.

Proof. Since

$$\mathbf{0}_V = \mathbf{0}_V + \mathbf{0}_V,$$

we have

$$f(\mathbf{0}_V) = f(\mathbf{0}_V + \mathbf{0}_V) = f(\mathbf{0}_V) + f(\mathbf{0}_V) \implies f(\mathbf{0}_V) = \mathbf{0}_W.$$

□

Theorem 4.3.2. (Linear maps are determined by their action on a basis)

Let $f : V \rightarrow W$ be a linear transformation, where V is finite-dimensional.

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis of V . Then f is determined by $f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)$.

More precisely, it means

1. $\forall \mathbf{v} \in V$, $f(\mathbf{v})$ is a linear combination of $f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)$.
2. Suppose the linear map $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$, $\forall i = 1, \dots, n$.
Then $g(\mathbf{v}) = f(\mathbf{v})$, $\forall \mathbf{v} \in V$.

Proof. Let $\mathbf{v} \in V$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis of V , we can write

$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$. Then

$$f(\mathbf{v}) = f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) = \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n).$$

In particular, $f(\mathbf{v})$ is determined by $f(\mathbf{v}_1), \dots, f(\mathbf{v}_n)$. This proves (1).

Now suppose another linear transformation $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$, then

$$\begin{aligned} g(\mathbf{v}) &= g(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= \alpha_1 g(\mathbf{v}_1) + \dots + \alpha_n g(\mathbf{v}_n) \\ &= \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n) \\ &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= f(\mathbf{v}). \end{aligned}$$

This proves (2)

□

1.4 Example

1.4.1 Rotation and Reflection

Let

$$\text{Rot}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

be defined by

$$\text{Rot}_\theta(\mathbf{v}) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \mathbf{v}.$$

Similarly,

$$\text{Ref}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

is defined by

$$\text{Ref}_\theta(\mathbf{v}) = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix} \mathbf{v},$$

which represents reflection along the line

$$y = \tan(\theta)x.$$

Since both functions are of the form

$$f(\mathbf{v}) = A\mathbf{v},$$

they satisfy the two properties of linear transformations.

Hence,

$$\text{Rot}_\theta, \text{Ref}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

are both linear transformations.

1.4.2 Differentiation

Define $D : \mathbb{R}[x]_{\leq n} \longrightarrow \mathbb{R}[x]_{\leq n-1}$ by

$$D(f(x)) = f'(x).$$

To verify that D is linear, check the two defining properties.

1. For any $f_1(x), f_2(x) \in \mathbb{R}[x]_{\leq n}$,

$$\begin{aligned} D(f_1(x) + f_2(x)) &= (f_1(x) + f_2(x))' \\ &= f_1'(x) + f_2'(x) \\ &= D(f_1(x)) + D(f_2(x)). \end{aligned}$$

2. For every $\lambda \in \mathbb{R}$,

$$\begin{aligned} D(\lambda f(x)) &= (\lambda f(x))' \\ &= \lambda f'(x) \\ &= \lambda D(f(x)). \end{aligned}$$

Therefore, D "behaves like matrices" $\implies D$ is a linear transformation.

1.4.3 Trace

Recall the trace map

$$\text{tr} : M_n(\mathbb{C}) \rightarrow \mathbb{C},$$

defined by

$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn},$$

where $A = (a_{ij})$.

To verify linearity,

1. For any $A, B \in M_n(\mathbb{C})$,

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B).$$

2. For every $\lambda \in \mathbb{C}$,

$$\text{tr}(\lambda A) = \lambda \text{tr}(A).$$

Hence,

$$\text{tr} : M_n(\mathbb{C}) \rightarrow \mathbb{C}$$

is a linear transformation.

1.4.4 Composition Operator

Fix a continuous function $g \in C(\mathbb{R})$. Then $S_g : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ is defined by

$$S_g(f(x)) = f(g(x)) \text{ is a linear transformation.}$$

In particular, if $g(x) = x - \alpha$ (with fixed $\alpha \in \mathbb{R}$), then S_g refers to the function that shifts the the graph of $f(x)$ by α units to the right.

Solution. To verify linearity,

1. For any $f_1, f_2 \in C(\mathbb{R})$,

$$\begin{aligned} S_g(f_1 + f_2) &= f_1(g(x)) + f_2(g(x)) \\ &= S_g(f_1) + S_g(f_2). \end{aligned}$$

2. For every $\lambda \in \mathbb{R}$,

$$\begin{aligned} S_g(\lambda f) &= \lambda f(g(x)) \\ &= \lambda S_g(f). \end{aligned}$$

Hence, $S_g : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ is a linear transformation.

□

2 Kernel and Image

2.1 Definition 4.4.1.

Let $f : V \rightarrow W$ be a linear transformation. Define

1. the **kernel** (or **nullspace**) of f by

$$\ker(f) = \{\mathbf{v} \in V \mid f(\mathbf{v}) = \mathbf{0}_W\}.$$

i.e. the kernel consists of all vectors in the domain that become the zero vector under f .

2. the **image** (or **range**) of f by

$$\operatorname{im}(f) = \{\mathbf{w} \in W \mid \mathbf{w} = f(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}.$$

i.e. the image/range consists of all the possible outputs of the function f .

2.2 Theorem

Theorem 4.4.1. (Properties of kernel)

Let $f : V \rightarrow W$ be a linear transformation.

1. $\ker(f)$ is a subspace of V .
2. f is injective if and only if $\ker(f) = \{\mathbf{0}_V\}$.

Proof. (1) To show that $\ker(f)$ is a subspace of V , we verify the three subspace conditions.

- Nonempty.

Since f is linear, $f(\mathbf{0}_V) = \mathbf{0}_W \implies \mathbf{0}_V \in \ker(f)$.

- Closed under addition.

Let $\mathbf{v}_1, \mathbf{v}_2 \in \ker(f) \implies f(\mathbf{v}_1) = \mathbf{0}_W, \quad f(\mathbf{v}_2) = \mathbf{0}_W$.

Check:

$$\begin{aligned} f(\mathbf{v}_1 + \mathbf{v}_2) &= f(\mathbf{v}_1) + f(\mathbf{v}_2) \\ &= \mathbf{0}_W + \mathbf{0}_W \\ &= \mathbf{0}_W. \end{aligned}$$

$$\implies \mathbf{v}_1 + \mathbf{v}_2 \in \ker(f).$$

- Closed under scalar multiplication.

Let $\lambda \in F$ and $\mathbf{v} \in \ker(f)$.

Then

$$\begin{aligned} f(\lambda \mathbf{v}) &= \lambda f(\mathbf{v}) \\ &= \lambda \mathbf{0}_W \\ &= \mathbf{0}_W. \end{aligned}$$

$$\implies \lambda \mathbf{v} \in \ker(f).$$

Therefore, $\ker(f)$ is a subspace of V .

(2) We show both directions.

($\Rightarrow$)

Suppose f is injective. If $\mathbf{v} \in \ker(f)$, then

$$f(\mathbf{v}) = \mathbf{0}_W = f(\mathbf{0}_V).$$

Since f is injective, $\mathbf{v} = \mathbf{0}_V \implies \ker(f) = \{\mathbf{0}_V\}$.

($\Leftarrow$)

Suppose $\ker(f) = \{\mathbf{0}_V\}$. If $f(\mathbf{v}_1) = f(\mathbf{v}_2)$, then

$$\begin{aligned} f(\mathbf{v}_1 - \mathbf{v}_2) &= f(\mathbf{v}_1) - f(\mathbf{v}_2) = \mathbf{0}_W. \\ \implies \mathbf{v}_1 - \mathbf{v}_2 &\in \ker(f) \end{aligned}$$

Therefore, $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}_V$, which implies $\mathbf{v}_1 = \mathbf{v}_2$.

Thus f is injective.

□

Theorem 4.4.2. (Properties of image)

Let $f : V \rightarrow W$ be a linear transformation.

1. $\text{im}(f)$ is a subspace of W .
2. f is surjective if and only if $\text{im}(f) = W$. ($\leftarrow$ def of surjectivity (nothing to prove))

Proof. (1) We verify the three subspace conditions.

- Nonempty.

Since $f(\mathbf{0}_V) = \mathbf{0}_W$, we have $\mathbf{0}_W \in \text{im}(f)$.

- Closed under addition.

Let $\mathbf{w}_1, \mathbf{w}_2 \in \text{im}(f)$.

Then there $\exists \mathbf{v}_1, \mathbf{v}_2 \in V$ s.t. $\mathbf{w}_1 = f(\mathbf{v}_1)$, $\mathbf{w}_2 = f(\mathbf{v}_2)$.

Hence,

$$\begin{aligned}\mathbf{w}_1 + \mathbf{w}_2 &= f(\mathbf{v}_1) + f(\mathbf{v}_2) \\ &= f(\mathbf{v}_1 + \mathbf{v}_2).\end{aligned}$$

Therefore, $\mathbf{w}_1 + \mathbf{w}_2 \in \text{im}(f)$.

- Closed under scalar multiplication.

Let $\mathbf{w} = f(\mathbf{v})$. Then $\lambda\mathbf{w} = \lambda f(\mathbf{v}) = f(\lambda\mathbf{v})$, so $\lambda\mathbf{w} \in \text{im}(f)$.

Therefore, $\text{im}(f)$ is a subspace of W .

(2)

By definition,

$$\begin{aligned}f \text{ is surjective} &\iff \forall \mathbf{w} \in W, \exists \mathbf{v} \in V \text{ s.t. } f(\mathbf{v}) = \mathbf{w} \\ &\iff W \subseteq \text{im}(f).\end{aligned}$$

Since $\text{im}(f) \subseteq W$ always holds, we conclude

$$f \text{ is surjective} \iff \text{im}(f) = W.$$

□

2.3 Example

2.3.1 Differentiation

Consider $D : \mathbb{R}[x]_{\leq 3} \longrightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x).$$

Solution. First, compute the kernel.

$$\begin{aligned}\ker(D) &= \{f \in \mathbb{R}[x]_{\leq 3} : D(f) = 0\} \\ &= \{f \in \mathbb{R}[x]_{\leq 3} : f'(x) = 0\} \\ &= \{\text{constant polynomials}\} \\ &= \mathbb{R}.\end{aligned}$$

As $\ker(D) \neq \{0\}$, we deduce that D is not injective.

Next, compute the image. Let

$$p(x) = ax^2 + bx + c \in \mathbb{R}[x]_{\leq 2}.$$

Then

$$p(x) = D\left(\frac{a}{3}x^3 + \frac{b}{2}x^2 + cx\right).$$

As $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$ (whole domain), D is surjective.

□

2.3.2 Trace

Consider $\text{tr} : M_2(\mathbb{R}) \longrightarrow \mathbb{R}$ defined by $\text{tr} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d$.

Solution. First, compute the kernel.

$$\begin{aligned} \ker(\text{tr}) &= \{A \in M_2(\mathbb{R}) : \text{tr}(A) = 0\} \\ &= \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}. \end{aligned}$$

Equivalently,

$$\ker(\text{tr}) = \text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}.$$

Hence, $\ker(\text{tr}) \neq \{0\}$, so tr is not injective.

Next, compute the image. For every $r \in \mathbb{R}$,

$$r = \text{tr} \begin{pmatrix} r & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus, $\text{im}(\text{tr}) = \mathbb{R} \implies \text{"tr" is surjective.}$

□

3 Rank-Nullity Theorem

3.1 Definition

Definition 4.5.1. Let $f : V \rightarrow W$ be a linear map. Define

1. the *nullity* of f by $\text{nullity}(f) = \dim(\ker(f))$.
2. the *rank* of f by $\text{rank}(f) = \dim(\text{im}(f))$

3.2 Theorem

Theorem 4.5.1. (Rank-Nullity Theorem)

Let V be a finite-dimensional vector space, and let $f : V \rightarrow W$ be a linear transformation. Then we have

$$\text{nullity}(f) + \text{rank}(f) = \dim(V).$$

Proof. Idea: $\ker(f) \subseteq V$

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$ (i.e. $\text{nullity}(f) = \dim(\ker(f))$)

By the Basis Extension Theorem, we may extend S to a basis of V :

$$S' = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\} : \text{a basis of } V. \text{ (i.e. } \dim(V) = n + k)$$

We claim that $S'' = \{f(\mathbf{s}_1), \dots, f(\mathbf{s}_k)\}$ forms a basis of $\text{im}(f)$.

Proof of Claim. We first show that it spans $\text{im}(f)$.

Let $\mathbf{w} \in \text{im}(f) \implies \mathbf{w} = f(\mathbf{v})$ for **some** $\mathbf{v} \in V$

As S' is a basis of V , we can write

$$\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i + \sum_{i=1}^k \beta_i \mathbf{s}_i.$$

Then,

$$\begin{aligned} \mathbf{w} &= f(\mathbf{v}) \\ &= f\left(\sum_{i=1}^n \alpha_i \mathbf{v}_i + \sum_{i=1}^k \beta_i \mathbf{s}_i\right) \\ &= \sum_{i=1}^n \alpha_i f(\mathbf{v}_i) + \sum_{i=1}^k \beta_i f(\mathbf{s}_i) \\ &= \sum_{i=1}^k \beta_i f(\mathbf{s}_i) \implies S'' \text{ spans } \text{im}(f) \end{aligned}$$

Next, we prove linear independence. Suppose

$$\sum_{i=1}^k r_i f(\mathbf{s}_i) = \mathbf{0}.$$

Then

$$f\left(\sum_{i=1}^k r_i \mathbf{s}_i\right) = \mathbf{0},$$

so

$$\sum_{i=1}^k r_i \mathbf{s}_i \in \ker(f).$$

As S is a basis of $\ker(f)$,

$$\sum_{i=1}^k r_i \mathbf{s}_i = \sum_{i=1}^n f_i \mathbf{v}_i \text{ for some scalars } f_i.$$

$$\implies \sum_{i=1}^k r_i \mathbf{s}_i + \sum_{i=1}^n (-f_i) \mathbf{v}_i = \mathbf{0}$$

As S' is a basis of V , it is linearly independent, this forces $r_i = 0, f_i = 0, \forall i$.

Thus, $\{f(\mathbf{s}_1), \dots, f(\mathbf{s}_k)\}$ is linearly independent. Hence it forms a basis of $\text{im}(f)$. □

Consequently,

$$\text{rank}(f) = \dim(\text{im}(f)) = \textcolor{red}{k} = (n + k) - n = \dim(V) - \text{nullity}(f),$$

□

3.3 Example

3.3.1 Differentiation

Recall the linear transformation $D : \mathbb{R}[x]_{\leq 3} \longrightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x).$$

We have $\ker(D) = \mathbb{R}$ and $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$.

Solution. clearly,

$$\text{nullity}(D) = \dim_{\mathbb{R}}(\ker(D)) = \dim_{\mathbb{R}}(\mathbb{R}) = 1$$

and

$$\text{rank}(D) = \dim_{\mathbb{R}}(\text{im}(D)) = \dim_{\mathbb{R}}(\mathbb{R}[x]_{\leq 2}) = 3$$

□

3.3.2 Trace

Recall that for $\text{tr} : M_2(\mathbb{R}) \longrightarrow \mathbb{R}$ defined by

$$\text{tr} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d,$$

we have $\ker(\text{tr}) = \text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$ and $\text{im}(\text{tr}) = \mathbb{R}$.

Solution. we have

$$\text{nullity}(\text{tr}) = \dim(\ker(\text{tr})) = 3,$$

and

$$\text{rank}(\text{tr}) = \dim(\text{im}(\text{tr})) = \dim(\mathbb{R}) = 1.$$

Notice that

$$3 + 1 = 4 = \underbrace{\dim(M_2(\mathbb{R}))}_{\text{domain}},$$

which equals the dimension of the domain. □

3.3.3 Identity Map

Let V be an n -dimensional vector space, and consider the identity map $\text{id}_V : V \rightarrow V$ defined by

$$\text{id}_V(\mathbf{v}) = \mathbf{v}, \quad \forall \mathbf{v} \in V.$$

Solution. Note that:

$$\ker(\text{id}_V) = \{\mathbf{0}_V\}, \quad \text{im}(\text{id}_V) = V,$$

we obtain

$$\text{nullity}(\text{id}_V) = \dim(\ker(\text{id}_V)) = \dim(\{\mathbf{0}_V\}) = 0,$$

and

$$\text{rank}(\text{id}_V) = \dim(\text{im}(\text{id}_V)) = \dim(V) = \text{dimension of the domain of } f = n.$$

Hence,

$$\text{rank}(\text{id}_V) + \text{nullity}(\text{id}_V) = n = \dim(V).$$

□

4 Rounding up

4.1 Remark

1. (Nullity and Rank of a Matrix)

A matrix $A \in M_{m \times n}(F)$ naturally induces the linear transformation $f_A : F^n \rightarrow F^m$ defined by $f_A(\mathbf{v}) = A\mathbf{v}$. From now on, we identify the matrix A with its associated linear transformation f_A . and denote f_A simply by A . That is:

$$\ker(A) = \ker(f_A) \text{ and } \operatorname{im}(A) = \operatorname{im}(f_A),$$

2. (Injectivity and Surjectivity)

For a linear transformation $f : V \rightarrow W$, we have

$$\operatorname{nullity}(f) = 0 \iff \ker(f) = \{\mathbf{0}_V\} \iff f \text{ is injective.}$$

Dually,

$$\operatorname{rank}(f) = \dim(W) \iff \operatorname{im}(f) = W \iff f \text{ is surjective.}$$